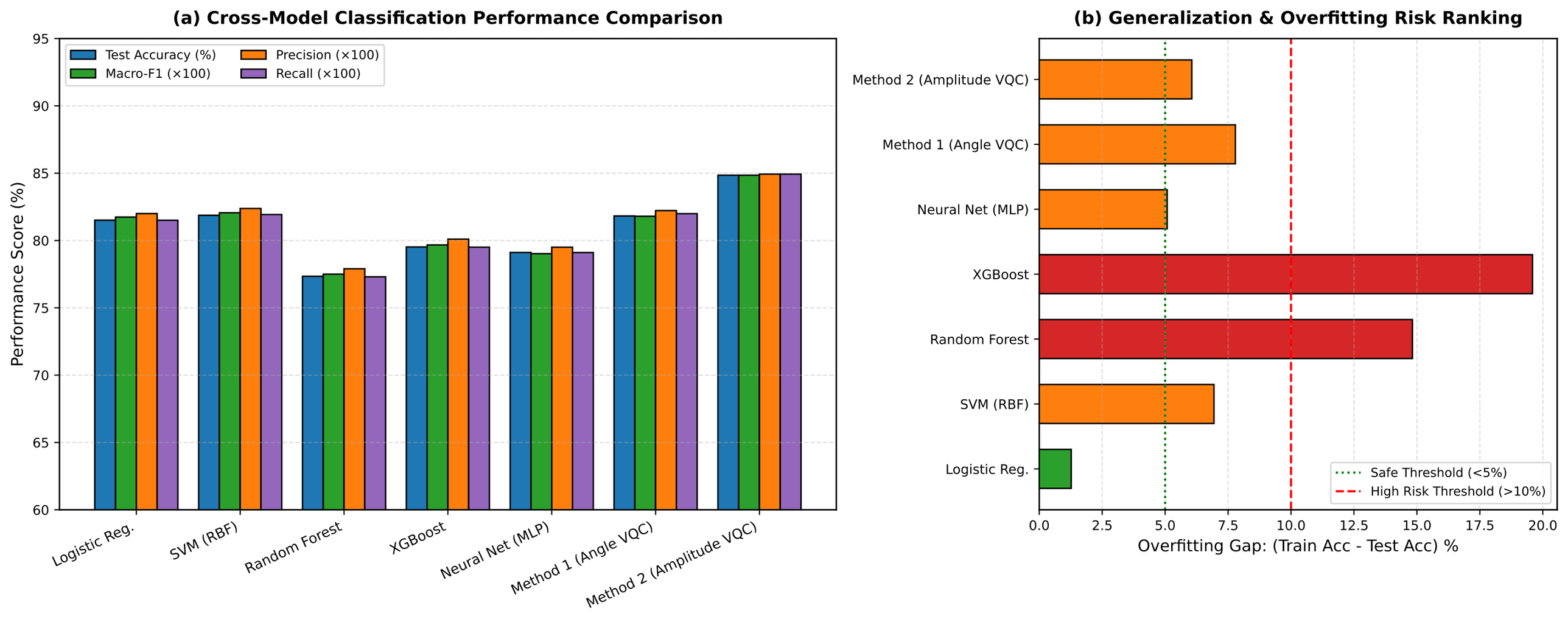
(a) Cross-Model Classification Performance Comparison
Test Accuracy (%)
Macro-F1 (×100)
Precision (×100)
Recall (×100)
Performance Score (%)
95
90
85
80
75
70
65
60
Logistic Reg.
SVM (RBF)
Random Forest
XGBoost
Neural Net (MLP)
Method 1 (Angle VQC)
Method 2 (Amplitude VQC)
(b) Generalization & Overfitting Risk Ranking
Method 2 (Amplitude VQC)
Method 1 (Angle VQC)
Neural Net (MLP)
XGBoost
Random Forest
SVM (RBF)
Logistic Reg.
Safe Threshold (<5%)
High Risk Threshold (>10%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Overfitting Gap: (Train Acc - Test Acc) %